

WASP-12b

R-1480

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_191211 · created 2026-08-01T12:50:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458828.81468 +/- 0.00068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.043
Transit depth (Rp/Rs)^2	1.47 +/- 1.04 [%]
Orbital Inclination (inc)	82.43 +/- 2.82
Airmass coefficient 1 (a1)	0.995 +/- 0.053
Airmass coefficient 2 (a2)	0.01 +/- 0.1
Scatter in the residuals of the lightcurve fit is	5.34 %
Best Comparison Star	#2 - [495, 69]
Optimal Aperture	2.51
Optimal Annulus	7.39
Transit Duration (day)	0.1216 +/- 0.0089

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.121 ± 0.043 vs 0.1178 (deviation 0.07σ)
Duration fit vs published	0.1216 d vs 0.125 d (deviation 0.38σ)
Mid-time O–C	-1.5 min
Depth SNR	2.8
Residual scatter	5.34 %

Light curve

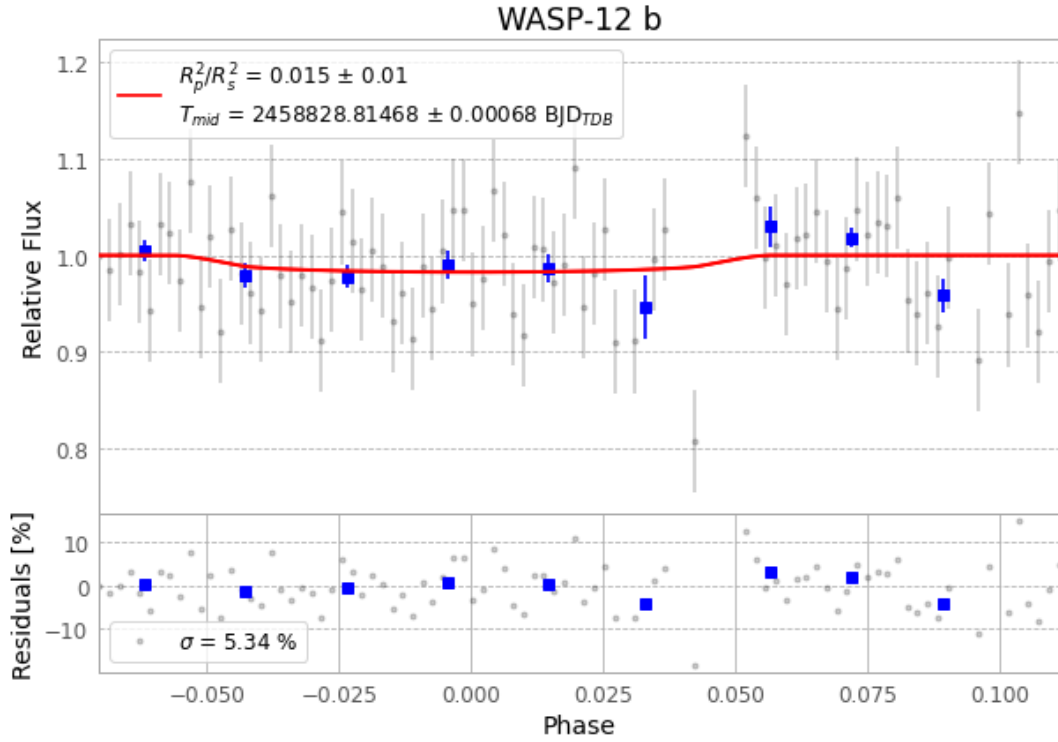


Figure 1 — FinalLightCurve_WASP-12 b_2019-12-10.png (SHA-256 10b62e3e751177f9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 252], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 1ca43249904125ce...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

96 FITS frames (63 MB), WASP-121912111053312.FITS → WASP-12191211101812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-191211.log (4152c6e9d6c2...), FinalLightCurve_WASP-12 b_2019-12-10.png (10b62e3e7511...), FinalLightCurve_WASP-12 b_2019-12-10.csv (39e80f115141...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 12:50Z.